Ex:No: 4B Date: 18.03.26	Decision Tree Algorithm Using Iris Dataset

Problem Description:

The Iris dataset contains measurements of iris flowers belonging to three species: *Setosa*, *Versicolor*, and *Virginica*. The goal is to use flower measurements such as sepal length, sepal width, petal length, and petal width to **classify the species of iris flowers** using a **Decision Tree algorithm**.

Objective:

To implement a Decision Tree classification model in R using the Iris dataset, perform train–test splitting, predict flower species, and evaluate model accuracy.

Procedure:

→ **Step 1: Install and Load Necessary Libraries**

- Install and load rpart (for decision tree modeling) and rpart.plot (for visualization).
- Load caret for dataset splitting and evaluation.

```
library (rpart)
library (rpart.plot)
library (caret)
```

- **Step 2: Load and Explore the Dataset**
  - Use the built-in iris dataset.
  - Print first six rows of iris dataset.

```
data (iris)
print ("First few rows of dataset")
print (head (iris))
```

OUTPUT:

[1] "First few rows of dataset :"

	Sepal.Length	Sepal.width	Petal.Length	Petal. width	Species
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
3	4.7	3.2	1.3	0.2	Setosa
4	4.6	3.1	1.5	0.2	Setosa
5	5.0	3.6	1.4	0.2	Setosa
6	5.4	3.9	1.7	0.4	Setosa

- **Step 3: Split the Dataset**
  - **80% for training, 20% for testing** using createDataPartition().

```
set.seed(123)
train_index <- createDataPartition(iris$Species, p=0.8,
                                   list=FALSE)
train_data <- iris[train_index,]
test_data <- iris[-train_index,]
```

- **Step 4: Train the Decision Tree Model**
  - Use the rpart() function to build the decision tree and print it.

```
tree_model <- rpart(Species~., data = train_data
                    method="class")
print(tree_model)
```

- **Step 5: Visualize the Decision Tree**
  - Use rpart.plot() to plot the decision tree structure.

```
rpart.plot(tree_model, main = "Decision Tree for
                                Iris Dataset",
type = 3, extra = 101, under = TRUE, tweak = 1.2,
box.palette = "RdBu")
```

OUTPUT:

n = 120

node), split, n, loss, yval, (yprob)

∴ denotes terminal node

1) root 120 75 virginica (0.33333333 0.21666667 0.37500000)

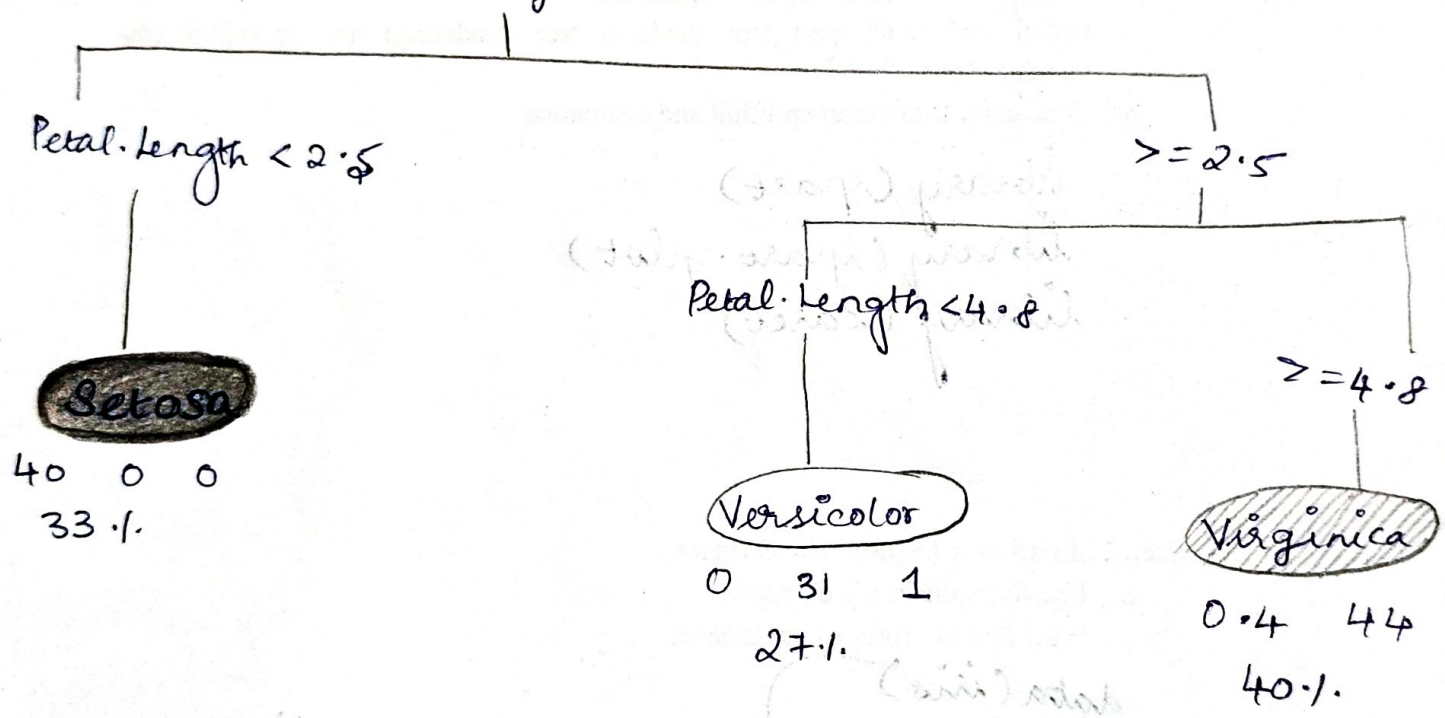
2) Petal. Length < 2.45 40 0 Setosa (1.00000000 0.00000000 0.00000000) ∴

3) Petal. Length >= 2.45 80 35 Virginica (0.00000000 0.43750000 0.56250000) ∴

4) Petal. Length < 4.75 32 1 versicolor (0.00000000 0.96875000 0.03125000) ∴

5) Petal. Length >= 4.75 48 4 Virginica (0.00000000 0.08333333 0.91666667) ∴

Decision Tree for Iris Dataset



- **Step 6: Evaluate the Model**
  - Predict on the test dataset.
  - Compute the **confusion matrix** and accuracy score.

Make Predictions on Test Data

```
predictions <- predict (tree_model, test_data
                        type = "class")
```

Evaluate Model Performance Confusion Matrix and Accuracy

```
conf_matrix <- confusionMatrix (predictions,
                                test_data$Species)
cat ("confusion Matrix : \n")
print (conf_matrix)
cat ("Accuracy of Decision Tree Mode : ",
conf_matrix$ overall [Accuracy], "\n")
```

Result:

The Decision Tree model was successfully implemented and visualized in R using the iris dataset. The model achieved an accuracy of 93.33%, showing high classification performance. The decision tree structure was visualized, helping in understanding how the algorithm makes predictions based on feature splits.

Confusion Matrix:

Confusion Matrix and Statistics

	References		
Prediction	setosa	versicolor	virginica
setosa	10	0	0
versicolor	0	13	0
Virginica	0	2	5

Overall statistics

Accuracy : 0.9333

95% CI : (0.7793, 0.9918

No Information Rate : 0.5

P-value [ACC > NIR] : 4.34 e-07

Kappa : 0.8947

Mcnemar's Test p-Value : NA

Statistics by class:

	class: setosa	class: versicolor	class: Virginica
sensitivity	1.0000	0.8667	1.0000
specifically	1.0000	1.0000	0.9200
pos pred value	1.0000	1.0000	0.7143
Neg pred value	1.0000	0.8824	1.0000
Prevalence	0.3333	0.5000	0.1667
Detection Rate	0.3333	0.4333	0.1667
Detection prevalence	0.3333	0.4333	0.2333
Balanced Accuracy	1.0000	0.9333	0.9600

Accuracy of Decision Tree Model : 0.9333333